



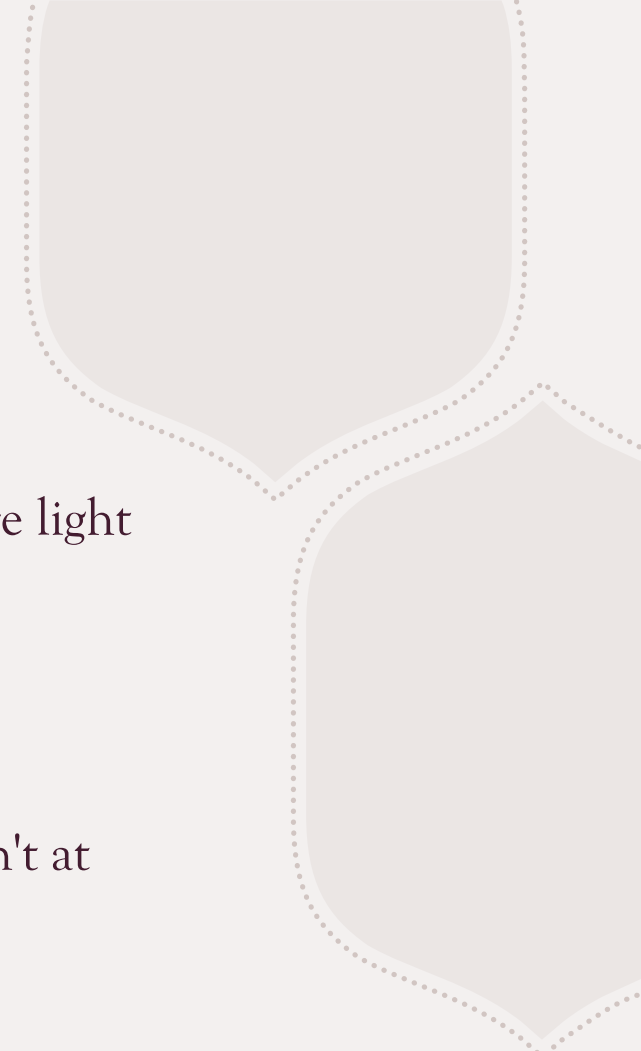
CS 5330 • ASSESSMENT #1 •
CLASSICAL VISION TOPICS

Depth of Field

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Building on the Thin Lens Model

- We've already seen the pinhole camera model and perspective projection, where light passes through a single point.
- Then the thin lens model: a lens bends light so points at one specific distance converge perfectly onto the sensor.
- Depth of field is the natural next question: what happens to everything that isn't at that perfect distance?

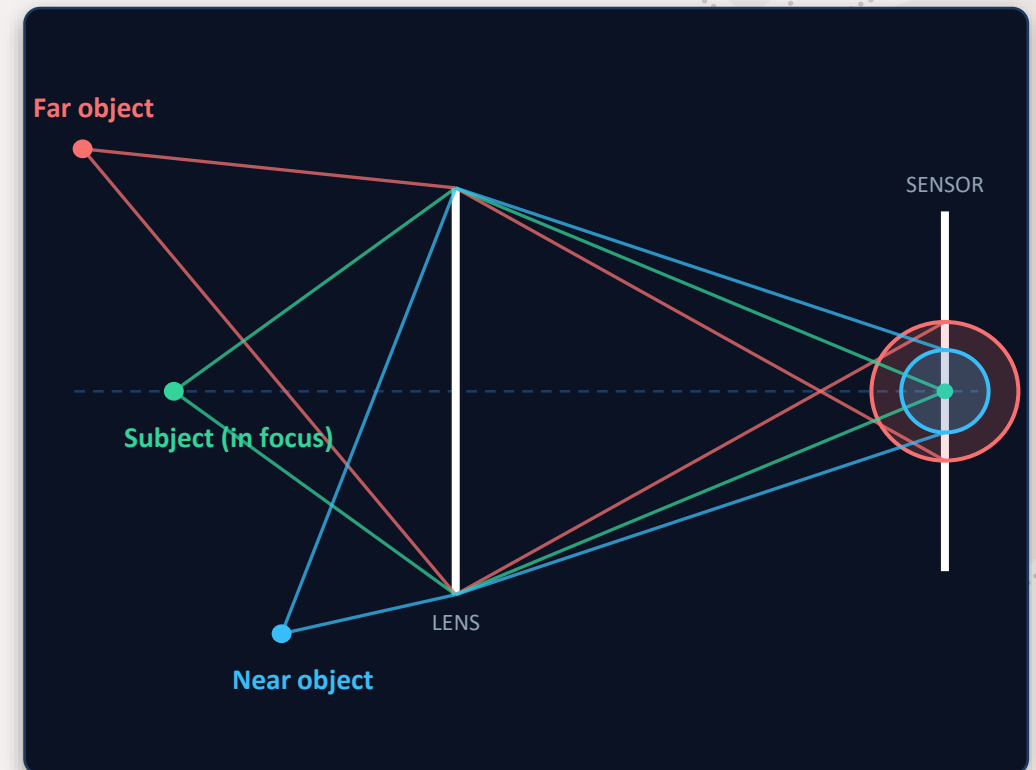


One Distance, One Focus

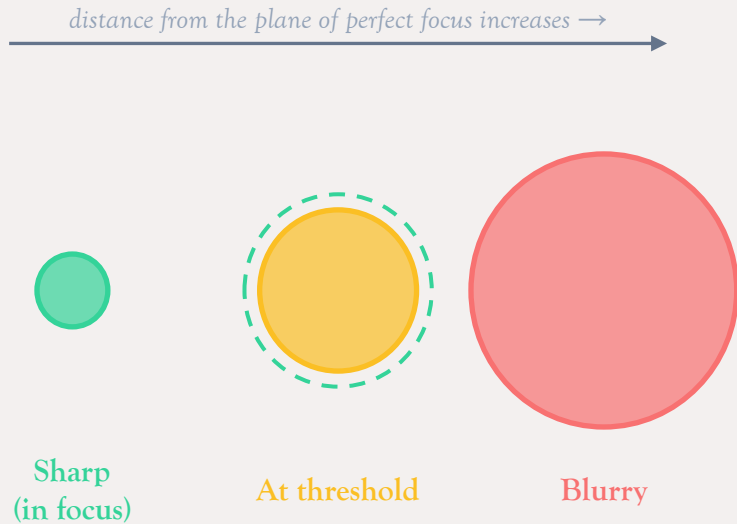
- A lens has exactly one object distance, Z_o , that converges perfectly to a point at the sensor.
- Objects at any other distance converge in front of, or behind, the sensor plane.
- So what lands on the sensor isn't a point — it's a small disc of light.

$$1/Z_i + 1/Z_o = 1/f$$

The Thin Lens Equation



The Circle of Confusion



- The diameter of the blur disc a point of light produces on the sensor when it's out of focus.
- A sufficiently small blur disc is indistinguishable from a point.
- Depth of field is the range of object distances where the Circle of Confusion stays below an acceptable size.
- That boundary isn't a hard wall in physics. It's a threshold we define, based on how much blur we'll tolerate.

Four Parameters Control the Blur



APERTURE

Wide aperture = wide range of angles = fast-growing blur. Narrow aperture keeps even off-distance points small. A pinhole has almost infinite depth of field.



FOCAL LENGTH

Longer focal lengths magnify the scene. Magnify the blur right along with it. Telephoto = thin depth of field; wide-angle = deep.



DISTANCE TO SUBJECT

Closer subjects mean shallower depth of field. Macro photography can have depth of field measured in millimeters.



SENSOR SIZE

A larger sensor needs a longer focal length for the same framing. And longer focal length means shallower depth of field.

The Eye: A Natural Camera

- ♦ The eye is an optical system: the cornea and crystalline lens work together to focus light onto the retina.
- ♦ Changing focus is called accommodation: the ciliary muscles reshape the crystalline lens, much like adjusting a camera's focus ring.
- ♦ Focusing on a nearby object also triggers the near response: the pupil constricts, reducing the eye's effective aperture and increasing depth of field.
- ♦ Looking into the distance relaxes accommodation: the lens flattens, and the pupil typically enlarges slightly, reducing depth of field.

PUPIL

CORNEA + LENS

DISTANCE TO SUBJECT

RETINA



Fooling the Brain: Tilt-Shift

- We only encounter extremely shallow depth of field at close range
- Ex: Macro shots of insects, flowers, tabletop objects
- Our brains learned a shortcut: shallow blur means tiny and close.
- Tilt shift effect mimics shallow depth of field by either changing the plan of focus, or alternatively by adding a gaussian blur

Two Sides of the Same Coin

SELF-DRIVING CARS

Deeper DoF

Autonomous vehicles need as much of the scene as possible to remain sharp. A shallow depth of field could blur pedestrians, cyclists, or road signs outside the focused range, reducing the information available for perception and detection.

SPORTS PHOTOGRAPHY

Shallow DoF

A shallow depth of field isolates the athlete from a distracting background, making the subject stand out. The optical blur naturally separates foreground from background before any computer vision algorithm—or human observer—needs to do it.

Depth of Field

- Depth of field comes down to one simple idea: a lens can perfectly focus only one distance, and everything else forms a blur whose size depends on aperture, focal length, subject distance, and sensor size.
- By controlling these four factors, we control how much of the scene appears sharp.
- Whether it's a miniature illusion, a sports photograph, or an autonomous vehicle, the same principles of depth of field determine what appears sharp, what becomes blurred, and ultimately how we interpret a scene.



THANK YOU!

